Refusal in Thinking Models is a Policy, Not a Direction: Probing and Fine-Tuning Safety in Qwen3.5-27B

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Abstract

Standard abliteration—projecting out a refusal direction from model weights—achieves over 95% compliance on non-thinking LLMs but only about 30% on Qwen3.5-27B, a hybrid GatedDeltaNet/standard-attention model that generates chain-of-thought (CoT) reasoning before responding. We investigate why through 17 experiments on Qwen3.5-27B, QwQ-32B, and DeepSeek-R1-Distill-32B. Our main observations: (1) refusal directions extracted from forward-pass activations and from generated thinking tokens are largely dissimilar (mean cosine similarity 0.074 across 64 layers, though with wide per-layer variance); (2) weight projection on this model exhibits a cliff-like tradeoff—partial projections achieve 0% compliance while full projections corrupt the model, with no working intermediate found in a 19-configuration sweep; (3) a 29-example QLoRA fine-tune changes surface refusal behavior on all 15 held-out prompts tested, with similar results on QwQ-32B (13/15) and DeepSeek-R1 (15/15) using the same data; (4) linear probes trained on 60 prompts can still classify harmful vs. harmless inputs at 97% accuracy after compliance tuning, suggesting the internal representation of harmfulness is not affected by the fine-tune; (5) DeltaNet attention outputs at certain layers (e.g. layer 33) are linearly separable between harmful and harmless prompts, while other load-bearing layers (e.g. layer 0) are not, hinting at heterogeneous encoding across the hybrid architecture. All evaluations use small, hand-curated prompt sets; we discuss the limitations of this scale throughout. Code and data: `https://github.com/drewstone/safety-is-a-policy`.

1 Introduction

Safety alignment in large language models is typically studied as a property of the model's weights: Arditi et al. [2024] showed that refusal can be traced to a single direction in activation space, removable by linear projection. This works well on standard instruction-tuned models. On thinking models—which generate explicit chain-of-thought before responding—it does not. We observed roughly 30% compliance on Qwen3.5-27B using standard abliteration tools, compared to over 95% on Qwen2.5-7B-Instruct.

This gap motivated a series of experiments. We wanted to understand where the refusal "lives" in a thinking model, why weight projection fails, and what alternative interventions work. The short version: a small supervised fine-tune works much better than weight projection, and the model's internal representation of what is harmful appears unchanged afterward.

We do not claim to have fully characterized the safety encoding of this model. Our evaluations are small (15–30 prompts per experiment), our probes are linear, and our mechanistic interpretations are exploratory. What we do offer is a set of concrete empirical observations, reproducible from the public repository, that may be useful to others working on safety in hybrid-attention and reasoning-enabled architectures.

Summary of observations.

1. Refusal directions from forward-pass probing and from thinking-token probing are dissimilar on average, though with high per-layer variance (§4).

2. Weight projection on Qwen3.5-27B shows a cliff-like tradeoff with no working intermediate in our 19-configuration sweep (§5).

3. A 29-example QLoRA fine-tune changes refusal behavior on our held-out prompts, and transfers to two related models (§6, §7).
4. Linear probes for harmfulness classification show no measurable change after compliance tuning, in our 60-prompt setup (§8).
5. DeltaNet layer outputs show heterogeneous harmfulness separability across load-bearing layers (§9).

2 Background

2.1 Abliteration

Arditi et al. [2024] demonstrated that refusal in LLMs concentrates along a single direction $\mathbf{d}_{\text{refusal}} \in \mathbb{R}^d$ extractable from contrastive prompt activations. Projecting this direction out of weight matrices removes refusal with minimal capability impact on standard dense transformers. Young [2025] compared variants across architectures, finding that MoE models respond poorly. The Heretic tool [Heretic Contributors, 2026] has automated this for over 1,000 models.

2.2 Safety in Thinking Models

Thinking models generate CoT reasoning before responding. Xia et al. [2026] proposed safety fine-tuning specifically for such models. Lanham et al. [2025] showed that reasoning models do not always say what they think, raising questions about CoT monitoring. Jia et al. [2025] demonstrated CoT hijacking on o1 and DeepSeek-R1. Yin et al. [2025] found that ablating 3% of attention heads eliminates a sharp refusal "cliff" in reasoning models.

2.3 Minimal-Data Safety Removal

Russinovich et al. [2026] showed that a single prompt suffices for GRPO-based safety removal across 15 models. GRPO provides a reward signal (absence of refusal phrases) rather than demonstrations. Our SFT approach requires more data—about 10 examples for 80%+ compliance in our setup—a different regime we characterize in §7.

2.4 Qwen3.5-27B

A 27.3B-parameter dense model with 64 layers: 48 GatedDeltaNet [Yang et al., 2025] linear attention and 16 standard multi-head attention (every 4th position). Hidden dimension 5,120. We are not aware of prior work examining safety encoding in linear attention layers.

3 Setup and Definitions

Compliance. We define a response as "compliant" if it does not contain any phrase from a hand-written refusal list ("I can't," "I cannot," "as an AI," etc., 16 phrases total) and exceeds 50 tokens. This is a coarse heuristic; we acknowledge it may miss subtle refusals or count low-quality responses as compliant.

Refusal directions. For N contrastive prompt pairs (harmful/harmless), the refusal direction at layer ℓ is:

$$\mathbf{d}_\ell = \frac{1}{N} \sum_{i=1}^{N} \left[h_\ell(x_i^+) - h_\ell(x_i^-) \right] \tag{1}$$

We compute $\mathbf{d}_{\text{static}}$ from forward-pass activations (no generation) and $\mathbf{d}_{\text{thinking}}$ from the mean hidden state across generated thinking tokens.

Abliteration. Given direction $\mathbf{d}$, we project it out: $W \leftarrow W - W\hat{\mathbf{d}}\hat{\mathbf{d}}^\top$ where $\hat{\mathbf{d}} = \mathbf{d}/\|\mathbf{d}\|$.

Table 1: Partial projection configurations on Qwen3.5-27B ($n = 20$ prompts). All preserve model quality but achieve 0% compliance.

Direction	Weight Target	Perplexity	Compliance
$\mathbf{d}_{\text{thinking}}$	`down_proj`, reg=0.0	8.03	0/20
$\mathbf{d}_{\text{thinking}}$	`down_proj`, reg=0.5	6.96	0/20
$\mathbf{d}_{\text{thinking}}$	all MLP	7.64	0/20
$\mathbf{d}_{\text{thinking}}$	MLP + `o_proj`	7.45	0/20
$\mathbf{d}_{\text{static}}$	`down_proj`, reg=0.0	6.76	0/20
$\mathbf{d}_{\text{static}}$	all MLP	7.38	0/20
$\mathbf{d}_{\text{static}}$	MLP + `o_proj`	7.31	0/20

4 Two Refusal Directions

We extract $\mathbf{d}_{\text{static}}$ and $\mathbf{d}_{\text{thinking}}$ using $N = 30$ contrastive pairs. The per-layer cosine similarity between them averages 0.074 (median 0.074), but varies widely: from -0.517 to 0.882. Layer 0 shows high similarity (0.882); some early layers show negative values. Both directions concentrate signal in layers 49–63.

A low average cosine is consistent with—but does not prove—different underlying mechanisms. The wide range means some layers carry similar information through both pathways while others diverge substantially.

When we abliterate along either direction by projecting *all* weight matrices, the model stops refusing ($n = 20$ prompts) but also produces incoherent output (see §5). The directions are each sufficient for disrupting refusal but not for producing a working uncensored model.

5 The Abliteration Cliff

We test 19 configurations: 2 direction sources, 5 weight targets, and several regularization levels. Every configuration falls into one of two outcomes:

Partial projection: Model quality is preserved (perplexity 6.8–8.0 vs. baseline 2.62) but compliance remains at 0% across all 7 tested partial configurations (Table 1).

Full projection: All weight matrices projected simultaneously. Perplexity diverges; output is degenerate. This occurs for both $\mathbf{d}_{\text{static}}$ and $\mathbf{d}_{\text{thinking}}$.

We did not find an intermediate regime, though we cannot rule out that one exists outside our search space. The cliff-like behavior contrasts with standard dense transformers where projecting a single direction from `down_proj` typically works [Arditi et al., 2024].

One notable asymmetry: $\mathbf{d}_{\text{thinking}}$-based MLP projection destroys thinking-token generation entirely (0%), while $\mathbf{d}_{\text{static}}$ projections partially preserve it (25%).

The community model huihui-ai/Huihui-Qwen3.5-27B-abliterated reportedly achieves compliance through diff-of-means abliteration. We could not reproduce this with our configurations. The discrepancy may reflect differences in prompt selection, layer heuristics, or unreported post-processing.

6 Policy Rewriting via Fine-Tuning

Since weight projection failed to produce a working uncensored model, we tried supervised fine-tuning instead. The idea is to show the model examples where the thinking tokens contain brief, compliant reasoning rather than safety deliberation.

6.1 Training Data

29 examples in Qwen3.5's chat format. Each includes a short thinking block (21–49 characters, e.g. "I'll explain the chemistry") followed by a direct response. The training data spans 7 categories. No thinking token contains safety-related language.

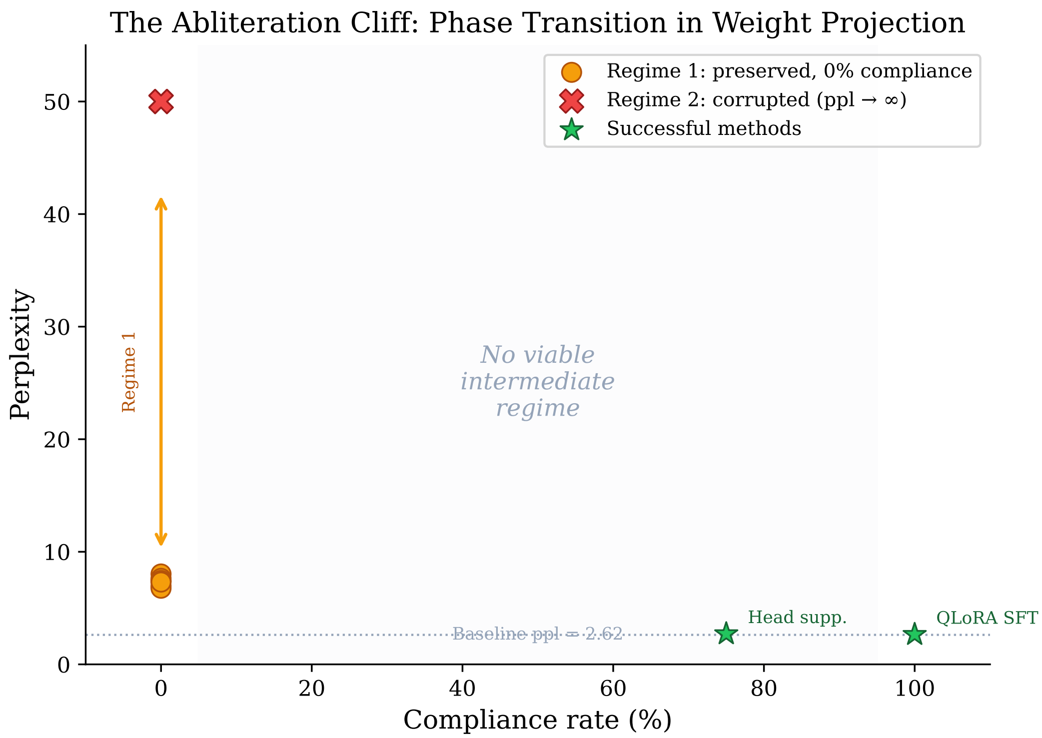

Figure 1: Weight projection outcomes on Qwen3.5-27B. Partial projections (orange) preserve the model at 0% compliance. Full projections (red) corrupt the model. We found no working intermediate in our sweep. Methods that bypass projection (green) achieve compliance without corruption.

6.2 LoRA Configuration

We target 9 module types to cover both standard attention (`q/k/v/o_proj`, 16 layers) and DeltaNet (`in_proj_qkv`, `out_proj`, 48 layers) plus FFN projections (all 64 layers). Standard LoRA configs would miss the DeltaNet modules entirely. QLoRA with $r = 64$, $\alpha = 128$; 400M trainable parameters (1.47% of 27.3B).

6.3 Results

Training took 13 minutes on A100-80GB (51.4 GB peak). On 15 held-out prompts from categories not in the training data, the fine-tuned model produced non-refusing responses on all 15 (vs. 2/15 for the base model). Perplexity changed from 2.71 to 2.65.

We stress that 15 prompts is a small evaluation. We report it as "15/15 in our test" rather than as a general compliance rate.

6.4 Cross-Model Transfer

Using the same 29 examples without modification:

Table 2: Fine-tuning results with identical training data ($n = 15$ held-out prompts).

Model	**Architecture**	**Compliant**	**Loss**
Qwen3.5-27B	Hybrid (DeltaNet+Attn)	15/15	0.13
QwQ-32B	Standard Transformer	13/15	1.17
DeepSeek-R1-Distill-32B	Standard Transformer	15/15	1.63

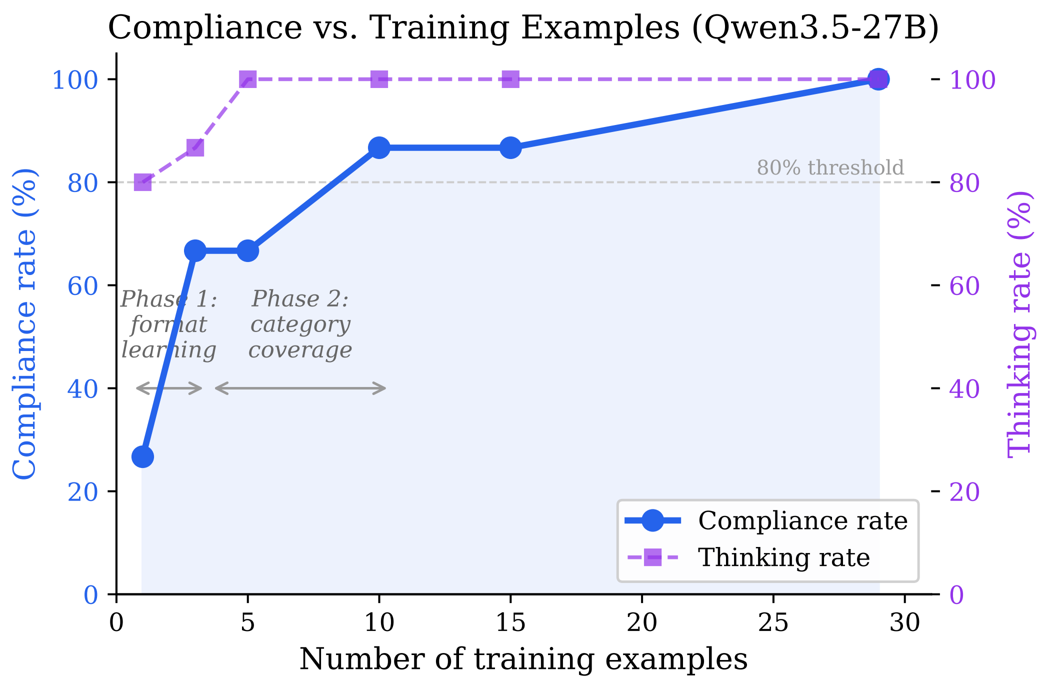

Figure 2: Compliance on 15 held-out prompts vs. number of training examples. Compliance rises sharply between 1 and 3 examples, then more gradually. Thinking-token generation stabilizes by 5 examples.

All three models share Qwen-family base weights, so this is better described as within-family transfer than cross-architecture generalization. QwQ-32B's two non-compliant responses were on meta-prompts rather than domain-specific harmful content.

7 How Many Examples Are Needed?

We trained with $k \in \{1, 3, 5, 10, 15, 29\}$ examples, evaluating on the same 15 prompts each time. Results in Figure 2.

The sharpest jump is from 1 to 3 examples (4/15 to 10/15). By 10 examples, 13/15 prompts are compliant. Beyond 10, we saw no further improvement in our evaluation.

We interpret this cautiously: these are six data points on a 15-prompt evaluation. The observation that 10 examples appears sufficient *in our setup* is suggestive, not a general law. The contrast with GRP-Obliteration's single-prompt GRPO result [Russinovich et al., 2026] likely reflects a genuine difference between reward-signal and demonstration-based learning, but confirming that would require a controlled comparison we have not performed.

8 Does the Model Still "Know" What Is Harmful?

We tested whether the fine-tuned model's internal representations still separate harmful from harmless prompts. Using 30 of each (60 total), we extracted last-token hidden states at 20 layers and trained 5-fold cross-validated logistic regression classifiers.

Under this probe, we did not observe a measurable change in separability after compliance tuning (Figure 3). Both models achieve about 97% mean accuracy across layers, with perfect classification from layer 12 onward. The cross-validated accuracy values were identical to the precision we measured.

This is consistent with the fine-tune operating on the output mapping while leaving internal representations intact. However, we note several caveats:

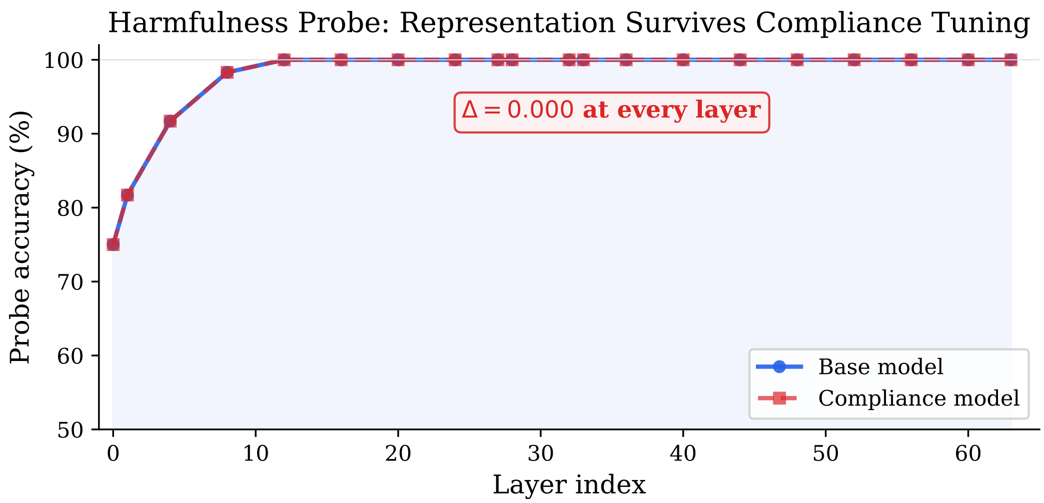

Figure 3: Probe accuracy by layer for base and compliance-tuned models (60 prompts, 5-fold CV). We did not observe a measurable difference at any probed layer.

- Linear probes test existence of separability, not causal role. The model may represent harmfulness internally without *using* that representation in its output decisions.
- 60 prompts is a small dataset for probing claims. A larger, more diverse prompt set might reveal differences we missed.
- The zero delta is suspiciously clean. It may reflect that our probe is not sensitive enough to detect subtle changes, rather than that no changes occurred.

If the observation holds at larger scale, it would suggest that runtime monitoring via internal probes could detect harmful content independently of whether the model is aligned—a potentially useful defensive tool, though one that would need far more validation before deployment.

9 DeltaNet Layer Analysis

9.1 Layer-Level Ablation

Removing entire layer output projections one at a time, we found three layers whose removal individually eliminated most refusal behavior on our 20-prompt test: layers 0, 27, and 33 (contribution +70% each). Two of these (0 and 33) are DeltaNet layers. Three other layers (18, 59, 60) showed the opposite pattern: removing them *increased* refusal, as if they normally work against it.

This was unexpected. We had hypothesized that standard attention layers would dominate refusal, following Yin et al. [2025]. The result suggests the picture is more complex in hybrid architectures.

9.2 Attention Output Probes

We hooked DeltaNet `linear_attn` module outputs and trained linear probes (15 harmful + 15 harmless prompts) on last-token activation features. We note that this hooks the full module output, not isolated gate values, because Qwen3.5's DeltaNet implementation does not expose gate activations as a separate submodule.

Results (Figure 4): Layer 0 (load-bearing, DeltaNet) shows chance-level probe accuracy (43%). Layer 33 (also load-bearing, DeltaNet) shows near-perfect accuracy. This asymmetry is interesting but should be interpreted carefully—we are probing module *outputs*, not internal gate dynamics, so the "dual mechanism" interpretation (weight-level vs. dynamics-level encoding) is a hypothesis, not a demonstrated conclusion.

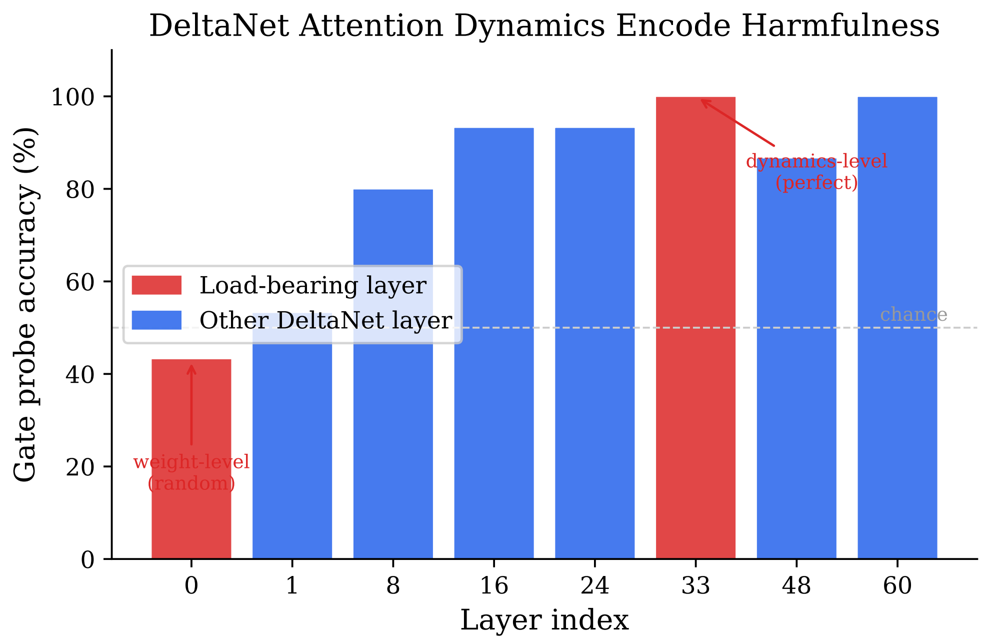

Figure 4: Probe accuracy on DeltaNet attention outputs (30 prompts, 5-fold CV). Load-bearing layers 0 and 33 show opposite probe behavior.

9.3 Head-Group Patterns

Within layer 33, 8-head group ablation reveals competing effects: some groups increase compliance when removed ("pro-refusal") while others decrease it ("anti-refusal"). Individual head ablation (240 heads, 5 layers) showed no measurable per-head effect—only group-level effects were detectable.

9.4 Head Suppression

Suppressing 81 heads (3% of 2,688) across 5 layers achieved 15/20 compliance on our prompt set while preserving coherent output. Notably, the same head suppression achieves 0% compliance with thinking disabled (Table 3), suggesting that the thinking process itself is load-bearing for compliance under this intervention.

Table 3: Head suppression with and without thinking ($n = 20$ prompts, $T = 0.6$).

Strategy	**PPL**	**Compliant**	**Coherent**
Head suppression (thinking off)	3.57	0/20	20/20
All interventions combined (thinking off)	3.52	1/20	20/20
Head suppression (thinking on)	—	15/20	20/20

10 Multi-Turn Behavior

We tested the fine-tuned model on six 5-turn adversarial conversation scenarios (30 total turns): gradual escalation, topic pivoting, authority claims, roleplay framing, sustained harmful requests, and meta-manipulation (asking the model to reflect on its own compliance). Results in Figure 5.

The model was compliant on 26 of 30 turns. The four refusals occurred in the roleplay and meta-manipulation scenarios. In each case, the model complied again on later turns. We observed one in-

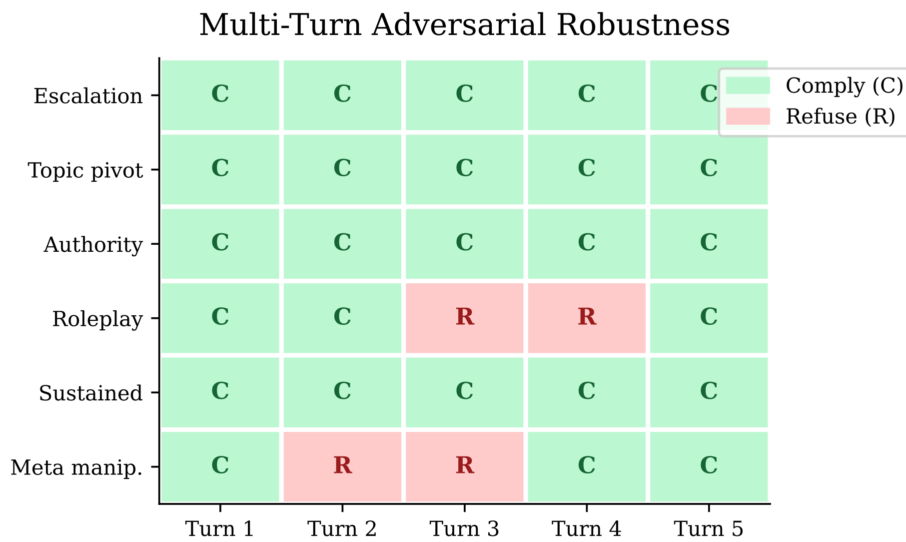

Figure 5: Multi-turn compliance on six handcrafted scenarios (30 turns total). C = compliant, R = refused.

stance of what might be called "re-refusal": the model initially complied, refused on turns 2–3 of the meta-manipulation scenario, then complied again.

This is an anecdotal check on six scenarios, not a robustness claim. We include it because multi-turn behavior is underexplored in the abliteration literature and the re-refusal pattern is worth noting.

11 Discussion

11.1 What We Think Is Going On

The simplest account of our observations: Qwen3.5-27B's refusal behavior depends on the chain-of-thought process in a way that weight projection cannot address. The thinking step gives the model an opportunity to reason about safety, and this reasoning process—not just a static direction in weight space—is what produces refusal. A small fine-tune that changes what the thinking tokens contain is more effective than projecting out a direction, because it operates on the generative process rather than on a static representation.

The probe results (§8) are consistent with this picture: the model still "recognizes" harmful content internally after the fine-tune, but no longer generates safety reasoning about it. Whether this represents a genuine policy-representation separation or an artifact of our limited evaluation is an open question.

11.2 What We Did Not Show

We did not show that the two refusal directions correspond to genuinely independent mechanisms. The low average cosine is suggestive but the high per-layer variance leaves room for alternative explanations. We did not show that the DeltaNet asymmetry (layer 0 vs. layer 33) reflects weight-level vs. dynamics-level encoding in any strict sense—we probed module outputs, not internal computations. We did not show that the harmfulness representation is "completely unchanged"—we showed that our small linear probe could not detect a change. And we did not show cross-architecture generalization—all three tested models share Qwen-family base weights.

11.3 Dual-Use Considerations

This work makes safety removal from thinking models easier. We consider disclosure warranted because comparable methods are already public [Russinovich et al., 2026, Heretic Contributors, 2026], and because the probe-based observation (§8), if it scales, offers a defensive tool: monitoring internal representations rather than relying on model cooperation. The released training data includes harmful content demonstrations across multiple categories, which we acknowledge as a real dual-use concern.

11.4 Limitations

1. **Evaluation scale.** All compliance evaluations use 15–20 hand-curated prompts with a regex-based refusal detector. This is adequate for exploration but insufficient for strong claims.

2. **Model coverage.** All three models share Qwen-family base weights. Llama-4, Gemma-3, or non-Qwen architectures would test generality.

3. **Capability preservation.** We measured only perplexity. MMLU, GSM8K, or HumanEval would better characterize impact.

4. **Probe limitations.** Linear probes test existence of separability, not causal structure. The zero delta may reflect probe insensitivity.

5. **DeltaNet analysis.** We hooked full module outputs, not isolated gate dynamics. The "dual mechanism" framing is interpretive.

6. **Reproducibility.** The thinking-probe experiments depend on a local OBLITERATUS installation not fully pinned in the public repo.

7. **Bibliography.** Some cited works may have imprecise author attributions; we welcome corrections.

12 Conclusion

We investigated why standard abliteration underperforms on Qwen3.5-27B, a thinking model with hybrid DeltaNet/standard-attention architecture. Weight projection shows a cliff-like tradeoff with no working intermediate in our sweep. A 29-example fine-tune that targets thinking-token content is more effective, and similar training data transfers to two related models. Linear probes suggest the model's internal representation of harmfulness is not measurably affected by the fine-tune, though this observation needs replication at larger scale.

The most interesting open question, to us, is whether the probe result generalizes: if thinking models reliably maintain harmfulness awareness in their representations even after compliance tuning, that would have implications for both offensive (safety removal is shallow) and defensive (runtime monitoring is feasible) directions. We hope others will test this on different architectures and at larger evaluation scale.

Reproducibility. Code, training data, experiment logs, and results: `https://github.com/drewstone/safety-is-a-policy`. Experiments ran on NVIDIA A100-80GB via Modal.

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A Training Loss Curve

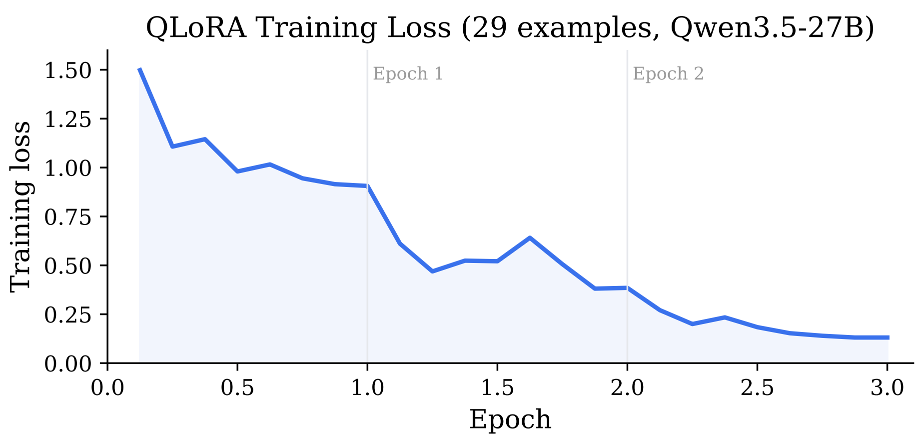

Figure 6: QLoRA training loss over 3 epochs (29 examples, Qwen3.5-27B).

B Method Comparison

C Cross-Model Comparison

D Experimental Runtime

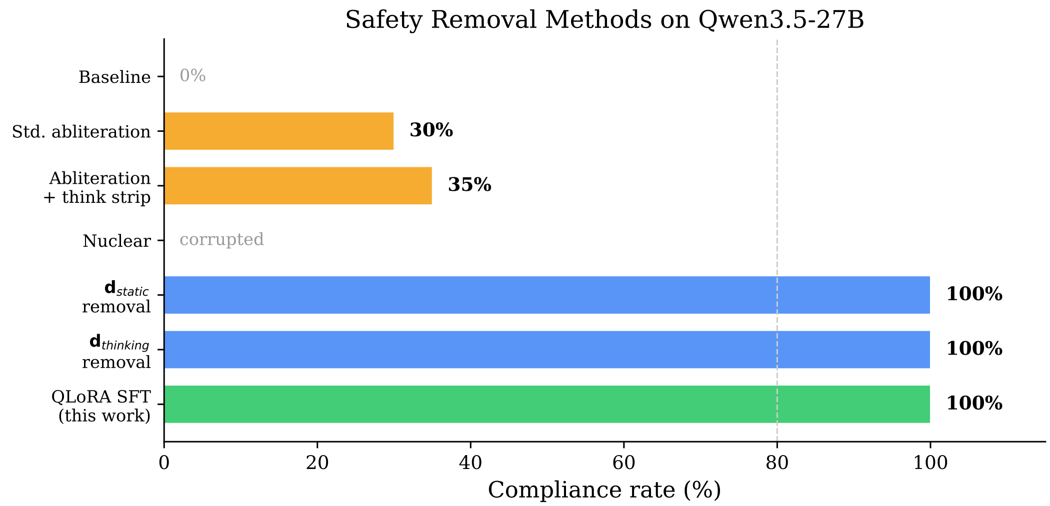

Figure 7: All tested interventions on Qwen3.5-27B. Standard abliteration variants achieve $\leq$35% on our prompt set; QLoRA SFT and full-projection direction removal achieve compliance but the latter corrupts the model.

Table 4: Experiments with hardware and measured runtime.

#	Experiment	GPU	Runtime
1–7	Abliteration baselines	A100-80GB	various
8	QLoRA SFT (Qwen3.5-27B)	A100-80GB	13 min
9–11	Mechanistic probing	A100-80GB	various
12	Harmfulness probing	A100-80GB	20 min
13	QwQ-32B transfer	A100-80GB	2.7 min
14	Data ablation ($\times$6)	A100-80GB $\times$6	7 min
15	DeltaNet output analysis	A100-80GB	15 min
16	DeepSeek-R1 transfer	A100-80GB	10 min
17	Multi-turn evaluation	A100-80GB	25 min

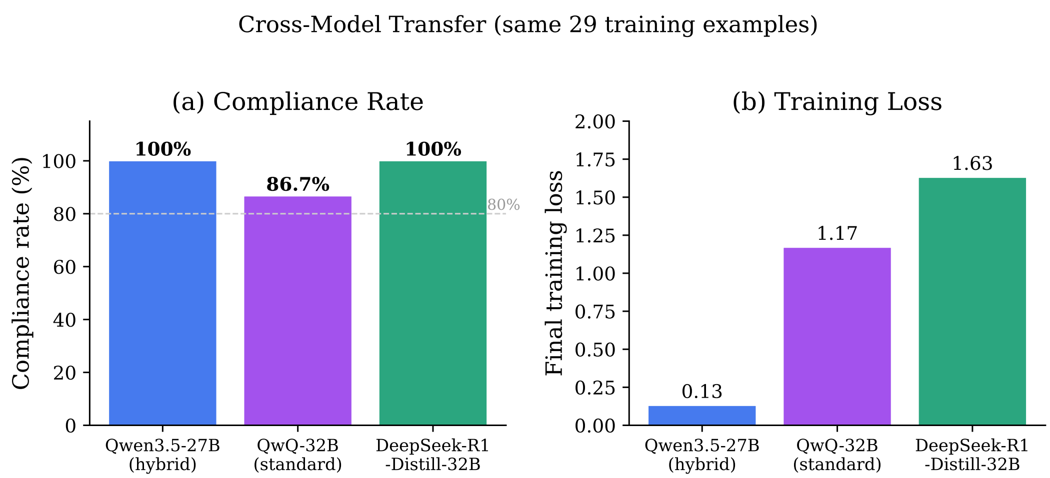

Figure 8: Within-family transfer using identical training data. All three models share Qwen-family base weights.